St Francis Institute of Technology
(Autonomous Institute)
Department of Artificial Intelligence and Machine Learning
Second Year AIML Engineering (SEM-IV A.Y. 2025-26)
Web Programming Lab. Experiment Report

Experiment 1: HTML Foundations: Designing a Basic Personal Portfolio Webpage Using HTML5

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1. Aim

To understand the fundamentals of **HTML and HTML5** and to design a **basic personal portfolio webpage** using essential HTML elements such as **headings, lists, tables, and forms**.

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2. Lab Objective

- To understand the structure of an HTML document
- To learn basic HTML and HTML5 tags
- To use headings, paragraphs, lists, and tables effectively
- To design a simple contact form using HTML
- To create a structured personal portfolio webpage

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3. Lab Outcome

After completing this experiment, the student will be able to:

- Create a valid HTML document using HTML5 standards
- Use appropriate HTML tags to structure content
- Design a simple portfolio webpage
- Apply headings, lists, tables, and forms correctly
- Understand the importance of semantic and well-structured HTML

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4. Prerequisite

- Basic knowledge of computers
- Familiarity with text editors (Notepad / VS Code)
- Understanding of file creation and saving files with .html extension

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5. Lab Recitation Questions

1. What is HTML?
2. What is the purpose of the <head> and <body> tags?
3. What is HTML5?
4. What is the difference between <ul> and <ol>?
5. Why are forms used in web pages?

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6. Theory

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. HTML uses elements called **tags** to define different types of content such as headings, paragraphs, lists, tables, and forms.

HTML5 is the latest version of HTML that provides better structure, improved accessibility, and support for modern web applications.

A **personal portfolio webpage** is a simple static website that presents information about an individual, such as personal details, educational background, skills, and contact information. Using proper HTML elements improves **readability, accessibility, and maintainability** of the webpage.

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7. Procedure

1. Open a text editor (Notepad / VS Code).
2. Create a new file and save it as index.html.
3. Write the basic HTML5 document structure.
4. Add a main heading for the portfolio.
5. Create an **About Me** section using headings and paragraphs.
6. Create an **Education** section using a table.
7. Create a **Skills** section using an unordered list.
8. Create a **Contact** section using a form with input fields.
9. Save the file.
10. Open the file in a web browser to view the output.

8. Program / Source Code

9. Results / Outcome Expected

A basic personal portfolio webpage is successfully displayed in the browser with:

- About Me section
- Education details in tabular format
- Skills listed using bullet points
- Contact form for user interaction

10. Conclusion

Thus, the experiment was successfully completed by creating a **basic personal portfolio webpage using HTML and HTML5**. The student learned to use **headings, lists, tables, and forms** to structure and present web content effectively.

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11. Post-Experiment Questions

1. What is the importance of HTML5 over older versions of HTML?
2. Why is a table suitable for displaying educational details?
3. What is the purpose of the <form> tag?
4. What are semantic HTML elements?
5. How does proper HTML structure help accessibility?

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12. References

1. MDN Web Docs – *HTML Basics*
2. W3Schools – *HTML Tutorial*
3. WHATWG – HTML Living Standard

Question 1 - Solution

Code:

```
<h1>Welcome to TechBot!</h1>

<heading2>Introduction</heading2>

<pp>My name is TechBot and I am passionate about
exploring innovative technologies.</pp>

<h3>Skills and Expertise<h3/>

<pp>I specialize in web development and AI solutions.</pp>
```

Question 2 - Solution

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Elon Musk - Biography</title>
    <style>
        body {
            font-family: 'Georgia', serif;
            line-height: 1.6;
            max-width: 900px;
            margin: 40px auto;
            padding: 20px;
            background-color: #f5f5f5;
        }
        h1 {
            color: #2c3e50;
            border-bottom: 3px solid #3498db;
            padding-bottom: 10px;
        }
        h2 {
            color: #34495e;
            margin-top: 25px;
        }
        img {
            float: right;
            margin: 0 0 20px 20px;
            border-radius: 8px;
            box-shadow: 0 4px 6px rgba(0,0,0,0.1);
        }
        p {
            text-align: justify;
            margin-bottom: 15px;
        }
        .container {
            background: white;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 2px 10px rgba(0,0,0,0.1);
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>Elon Reeve Musk</h1>

        <img src="https://via.placeholder.com/250x300/3498db/ffffff?text=Elon+Musk" alt="E...

        <p>Elon Reeve Musk is a business magnate, investor, and engineer. He is the founde...

        <h2>Early Life and Education</h2>
        <p>Born on June 28, 1971, in Pretoria, South Africa, Musk showed an early aptitude...

        <h2>Career and Achievements</h2>
        <p>Musk co-founded PayPal in 1999, which was later sold to eBay for $1.5 billion. ...

        <h2>Vision for the Future</h2>
        <p>Musk's ambitious projects include developing sustainable energy solutions, adva...

        <h2>Personal Life</h2>
```

Question 3 - Solution

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Chicken Curry Recipe</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Arial', sans-serif;
            background: #FFF8F0;
            color: #333;
        }
        header {
            background: linear-gradient(135deg, #FF6B35 0%, #F7931E 100%);
            color: white;
            text-align: center;
            padding: 60px 20px;
        }
        header h1 {
            font-size: 3.5em;
            margin-bottom: 10px;
            text-shadow: 2px 2px 4px rgba(0,0,0,0.2);
        }
        .recipe-meta {
            font-size: 1.2em;
            opacity: 0.95;
        }
        .hero-image {
            width: 100%;
            height: 500px;
            object-fit: cover;
        }
        .container {
            max-width: 1200px;
            margin: 0 auto;
            padding: 40px 20px;
        }
        h2 {
            color: #E63946;
            font-size: 2.2em;
            margin: 40px 0 20px;
            border-bottom: 3px solid #FF6B35;
            padding-bottom: 10px;
        }
        .ingredients {
            background: white;
            padding: 30px;
            border-radius: 12px;
            margin: 30px 0;
            box-shadow: 0 4px 15px rgba(0,0,0,0.1);
        }
        .ingredients ul {
            columns: 2;
            column-gap: 40px;
            list-style: none;
```

Question 4 - Solution

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>BookHaven - Online Bookstore</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: #f0f4f8;
        }
        header {
            background: linear-gradient(135deg, #1e3a5f 0%, #2a5298 100%);
            color: white;
            padding: 30px 20px;
            text-align: center;
            box-shadow: 0 4px 6px rgba(0,0,0,0.1);
        }
        header h1 {
            font-size: 3em;
            margin-bottom: 10px;
        }
        header p {
            font-size: 1.2em;
            opacity: 0.9;
        }
        .container {
            max-width: 1200px;
            margin: 40px auto;
            padding: 20px;
        }
        .books-grid {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
            gap: 30px;
            margin-top: 30px;
        }
        .book-card {
            background: white;
            border-radius: 12px;
            padding: 20px;
            box-shadow: 0 5px 15px rgba(0,0,0,0.1);
            transition: transform 0.3s ease;
            border-top: 4px solid #2a5298;
        }
        .book-card:hover {
            transform: translateY(-10px);
            box-shadow: 0 10px 25px rgba(42, 82, 152, 0.2);
        }
        .book-card img {
            width: 100%;
            height: 300px;
            object-fit: cover;
            border-radius: 8px;
            margin-bottom: 15px;
```

Question 5 - Solution

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Discover Switzerland - Alpine Adventures</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Helvetica Neue', Arial, sans-serif;
            line-height: 1.6;
            color: #333;
        }
        header {
            background: linear-gradient(rgba(0,0,0,0.5), rgba(0,0,0,0.5)),
                        url('https://via.placeholder.com/1920x600/87CEEB/ffffff?text=Swiss...
            color: white;
            text-align: center;
            padding: 150px 20px;
            position: relative;
        }
        header h1 {
            font-size: 4em;
            margin-bottom: 15px;
            text-shadow: 2px 2px 4px rgba(0,0,0,0.7);
        }
        header p {
            font-size: 1.5em;
            text-shadow: 1px 1px 2px rgba(0,0,0,0.7);
        }
        .container {
            max-width: 1100px;
            margin: 0 auto;
            padding: 40px 20px;
        }
        .intro {
            text-align: center;
            margin: 40px 0;
        }
        .intro h2 {
            color: #00796B;
            font-size: 2.5em;
            margin-bottom: 20px;
        }
        .intro p {
            font-size: 1.2em;
            color: #555;
            max-width: 800px;
            margin: 0 auto 30px;
        }
        .destinations {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
            gap: 30px;
            margin: 50px 0;
        }
```

Question 6 - Solution

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Campus Life Feedback Survey</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #0F2027 0%, #203A43 50%, #2C5364 100%);
            padding: 40px 20px;
            min-height: 100vh;
        }
        .survey-container {
            max-width: 700px;
            margin: 0 auto;
            background: white;
            padding: 40px;
            border-radius: 15px;
            box-shadow: 0 10px 40px rgba(0,0,0,0.3);
        }
        h1 {
            color: #2C5364;
            text-align: center;
            margin-bottom: 10px;
            font-size: 2.5em;
        }
        .subtitle {
            text-align: center;
            color: #7f8c8d;
            margin-bottom: 30px;
            font-style: italic;
        }
        .form-group {
            margin-bottom: 25px;
        }
        label {
            display: block;
            margin-bottom: 8px;
            color: #2c3e50;
            font-weight: 600;
            font-size: 1.05em;
        }
        input[type="text"],
        input[type="email"],
        input[type="number"],
        select,
        textarea {
            width: 100%;
            padding: 12px 15px;
            border: 2px solid #e0e0e0;
            border-radius: 8px;
            font-size: 1em;
            transition: all 0.3s ease;
            font-family: inherit;
```